



BEST PRACTICE SERIES

**Utilizing VIMS Data for Effective
Planned Maintenance**

Application Maintenance Site Management Component Rebuild Safety MARC Management

Utilizing VIMS Data for Effective Planned Maintenance0

1.0 Introduction.....1

2.0 Best Practice Description.....1

3.0 Implementation Steps3

4.0 Benefits.....3

5.0 Resources Required4

6.0 Supporting Attachments / References4

7.0 Related Best Practices4

8.0 Acknowledgements.....4

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

1.0 Introduction

This Best Practice illustrates how to use Vital Information Management System (VIMS) data for effective and cost-efficient Planned Maintenance (PM). PM events will better contribute to optimum machine productivity when the actual downtime associated with the PM is curtailed.

Downtime (associated with PM events) must be minimized, and all necessary work completed, to effectively increase machine productivity. Successful PM events also require reviewing VIMS data shortly after the machine leaves the shop. Reviewing the data afterward will ensure the proper maintenance issues were corrected, and further help to reduce unscheduled downtime.

2.0 Best Practice Description

1. Three days prior to PM
 - a. Download VIMS data, using VIMSpC,
 - b. Take oil samples
 - c. Review backlog
 - d. Mechanical inspection
 - e. Analyze the VIMS data and complete the VIMS PM Planner **(see Figure 1 on next page)**
 - f. Schedule the proper number of people required to complete the PM
 - g. Order and receive the required parts to fix any mechanical issues discovered.
 - h. Establish an overall plan to complete the PM in the most effective manner
2. One day after PM.
 - a. Download the VIMS data and fill out the PM Planner again
 - b. Ensure all maintenance completed was accurate. (This can also serve as a training opportunity for newer technicians.)

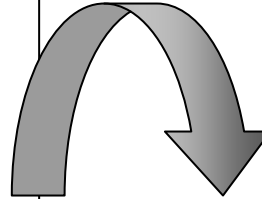
THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Utilizing VIMS Data for Effective Planned Maintenance	DATE 1/12/2015	CHG NO 03	NUMBER 1006-2.15-1027
---	-------------------	-----------------	--------------------------

From: 6/5/2003 3:14:13 PM				To: 7/7/2003 9:02:00 AM		
Event ID	Description	Value	Level	Max Dur.	Total	Hrs/Evt
Body Up;Gnd Spd				Total Count: 6		
1516	Body Up;Gnd Spd	8.39mph	2	0-0:1:0	6	33.58
Engine Derate				Total Count: 1		
1719	Engine Derate	6. %	1	0-3:26:14	1	201.49
Fuel Lvl				Total Count: 5		
83	Fuel Lvl	6.5%	2	0-0:3:34	2	100.74
84	Fuel Lvl	49-96-3	2	0-0:43:5	1	201.49
1227	Fuel Lvl	49-96-4	2	0-0:1:48	2	100.74
Lo Strg Pres				Total Count: 1		
174	Lo Strg Pres	Low	3	0-0:0:6	1	201.49
ParkBk				Total Count: 4		
292	ParkBk	27-627-4	2	0-1:0:23	4	50.37
Rt Exh Temp				Total Count: 31		
1422	Rt Exh Temp	1443.2°F	2	0-0:0:9	31	6.50
Strg Oil Temp				Total Count: 3		
959	Strg Oil Temp	224.6°F	1	0-0:49:24	3	67.16
TC Out Temp				Total Count: 2		
1063	TC Out Temp	264.2°F	2	0-0:1:7		
VIMS Snapshot				Total Count: 2		
602	VIMS Snapshot	0. %	1	ACTIVE		
1147	VIMS Snapshot	0. %	1	ACTIVE		

Figure 1:

VIMS Data transferred into VIMS PM Planner



CATERPILLAR®		VIMS PM Planner			
Data Events					
No	Description	Comments			
1.	Engine Derate	6%, Level 1			
2.	Lo Strg Pres	Low, Level 3			
3.	Rt Exh Temp	1443.2°F, Level 2			
4.	TC Out Temp	264.2 °F, Level 2			
System Events					
No	Description	MID	CID	FMI	Comments
1.	Fuel Lvl	49	96	3 and 4	3 Events. FMI 3, 4. One had value of 8.5%
2.	Park Bk	27	627	4	
Trends					
No	Description	Condition	Units	Avg	App. Guide
1.	Rt Lt Exh Temp	Engine Loaded	°F	244.64	+/- 90
2.	Air Filter	Engine Loaded	in H ₂ O	15.55	20
3.	Boost Pressure	Engine Loaded	psi	21.09	
4.	Engine Oil Pressure	Engine High & Cool > 74	psi	63.8	
5.	Engine Oil Pressure	Engine Low & Cool > 75	psi	61.92	
6.	RtF-LtF Susp cyl	Travelling Empty	psi	-71.66	+/- 55
7.	RtR-LtR Susp cyl	Travelling Empty	psi	21.53	+/- 55
8.	RtF-LtF Brake Temp	Travelling	°F	0.01	+/- 4
9.	RtR-LtR Brake Temp	Travelling	°F	-0.01	+/- 4
10.	TC LUC Upshift		s	1.16	1.10-1.125
11.	TC LUC Downshift		s	1.22	1.10-1.50
12.	Transmission Slip	Gear Change: 1-2	s	N/A	N/A
13.	Transmission Slip	Gear Change: 2-3	s	0.7	0.625-0.700
14.	Transmission Slip	Gear Change: 3-4	s	0.63	0.55-0.65
15.	Transmission Slip	Gear Change: 4-5	s	0.66	0.60-0.75
16.	Transmission Slip	Gear Change: 5-6	s	0.64	0.575-0.65
Additional Comments or Recommendations:					

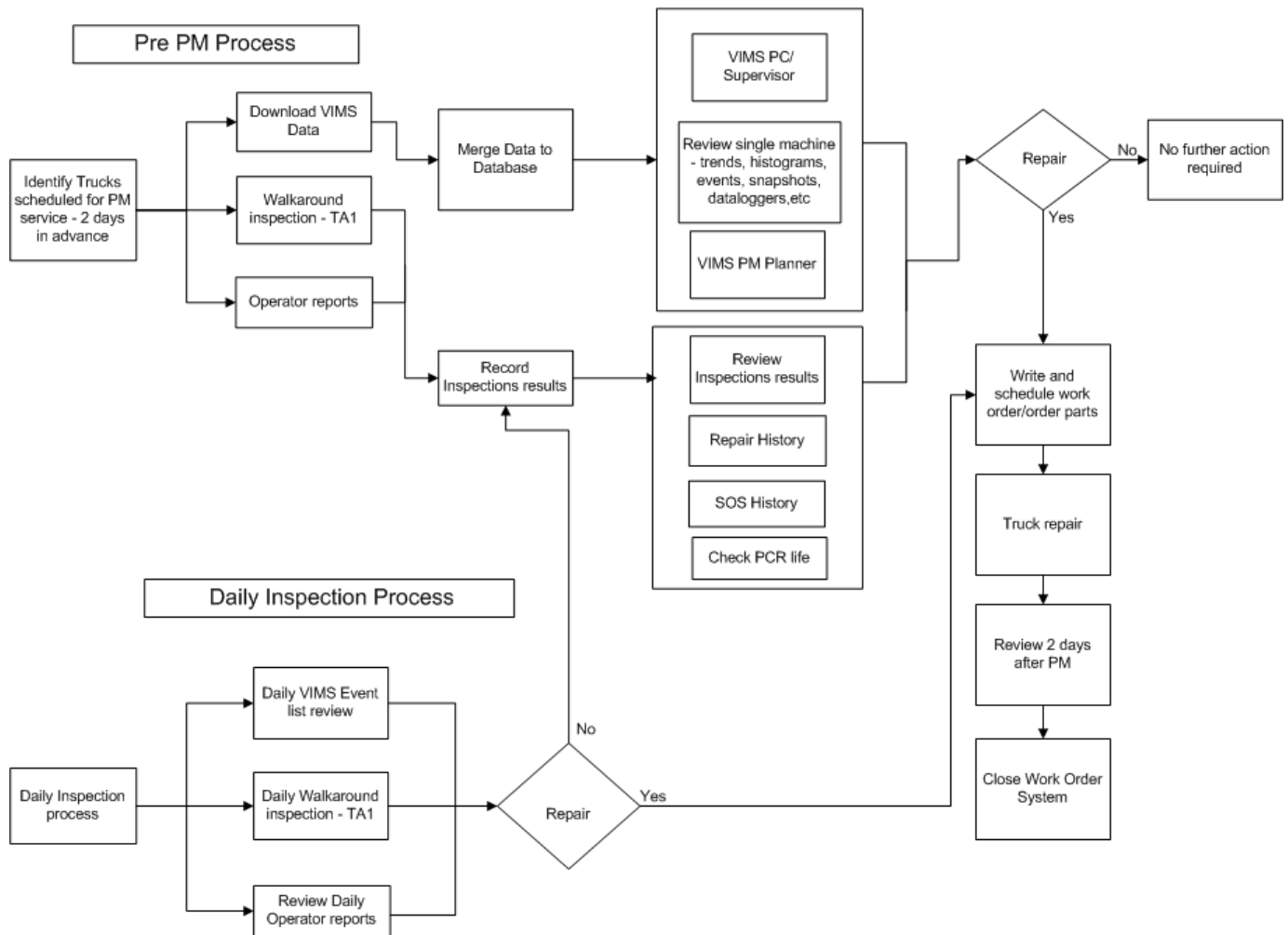
THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Utilizing VIMS Data for Effective Planned Maintenance

DATE
1/12/2015CHG
NO
03NUMBER
1006-2.15-1027

VIMS PM Process

September, 2006



3.0 Implementation Steps

Three P's are the key steps to getting started:

- People** – Having the proper people dedicated to gathering, analyzing, and reporting the finding of the VIMS data to the maintenance manager.
- Parts** – Ordering and having the needed parts on hand when the machine enters the shop.
- Plan** – Having an over all plan to schedule enough people and time to complete the PM in the most efficient manner.

4.0 Benefits

The benefits of utilizing the VIMS PM planner are:

- The ability to complete maintenance needs during a scheduled downtime.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Utilizing VIMS Data for Effective Planned Maintenance

DATE
1/12/2015

CHG
NO
03

NUMBER
1006-2.15-1027

- The ability to save time and resources by having a plan prior to the PM.
- Taking a proactive maintenance approach to help improve machine availability.
- Can serve as a training opportunity for newer technicians to ensure quality maintenance.

5.0 Resources Required

People dedicated to gathering and analyzing VIMS data, are the primary resource.

VIMS Application training may also be required to assist with the data collection step of the process, using the *VIMSpC* software.

6.0 Supporting Attachments / References

Attachments:

- [*VIMS PM Planner.doc*](#) (Electronic Microsoft-Word Document can be found within the "Attachments" tab of electronic Adobe file.)
- [*VIMS PM Planner.ppt*](#) (Electronic Microsoft-PowerPoint file can be found within the "Attachments" tab of electronic Adobe file.)
 - Provides an overview of how to utilize the PM Planner Template
- [*VIMS Process & Daily Inspection Flowdem.vsd*](#) (Electronic Microsoft-Visio file can be found within the "Attachments" tab of electronic Adobe file.)

References:

- VIMSpC Subscription numbers:
 - JERD2137 (Dealer Site License)
 - JERD2138 (Dealer Annual Subscription)
 - JERD2175 (Customer Annual Subscription)
- VIMS Application Guide (SELD7001)

7.0 Related Best Practices

None applicable at this time.

8.0 Acknowledgements

This Best practice was written by:

Justin C. Swisher
Caterpillar Global Mining
Product Evaluation Engineer
swisher_justin_c @cat.com
1-309-494-4169

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Utilizing VIMS Data for Effective Planned Maintenance	DATE 1/12/2015	CHG NO 03	NUMBER 1006-2.15-1027
---	-------------------	-----------------	--------------------------